UNITED INTERNATIONAL UNIVERSITY

V1.3 / 2026

URRT

URC

URRT ROBO CUP

MECHANICAL & ELECTRICAL

01

MOBILITY

Rails slopes gravel and narrow paths

02

DEXTERITY

QR codes touch targets box smash and the sensor crate

03

AUTONOMY

Hands-off execution worth far more points

rescuerover.uiu.ac.bd • rescuerover@cair.uiu.ac.bd

#URRT • #URRT • #URRT • #URRT • #URRT • #URRT •

DOCUMENT SET

This is the **Mechanical & Electrical Rulebook**, part of the four-document URRT Robo Cup set:

- **General Rulebook** – format, milestones, arena, tasks, scoring including the Build Quality Score, run procedure, penalties, ranking.
- **Mechanical & Electrical Rulebook** – rover size, mechanical design and CAD, electrical and wiring rules, E-stop, power, communications, classes, safety.
- **Software & Autonomous Rulebook** – control modes, autonomy, sensor crate, dashboard.
- **Judgement Sheet** – inspection, score, dashboard and build quality sheets, milestone form, judge quick reference.

Where a rule lives in another document, this book points to it by name.

1. ROVER SPECIFICATIONS

1.1 Size and Configuration

Parameter	Limit	Notes
Maximum length	300 mm	Measured in the starting configuration.
Maximum width	300 mm	Flippers and sensors stowed.
Maximum height	300 mm	Antennas and masts must be folded or stowed.

Table 1. Rover size limits.

1.1.1 The rover must fit inside a $300 \times 300 \times 300$ mm gauge in its starting configuration. Parts may be deployed or extended *after* the start command, provided they were stowed at inspection.

1.1.2 Rovers shall not carry an arm, gripper or any other manipulator. All interaction with the arena is done with the chassis, the flippers (where fitted) and the on-board sensors.

1.1.3 A rover that exceeds the size limit is not permitted to start until it complies. There is no mass limit.

1.1.4 A rover must keep the same mechanical configuration and class for the whole championship unless the Technical Committee approves a change in writing.

1.2 Emergency Stop

1.2.1 Every rover shall have a clearly visible, red, mushroom-head emergency stop (E-stop) button that is easy to reach from outside the rover without crossing any moving part.

1.2.2 Pressing the E-stop shall remove power from all actuators (drive motors and flippers). The function shall be implemented in hardware and shall not depend on software or on the wireless link.

1.2.3 The E-stop must be tested at inspection and may be tested by referees at any time. A rover that fails the E-stop test is not permitted to run.

1.2.4 Releasing the E-stop must not restart any motion. A deliberate restart action is required.

1.3 Communications

1.3.1 Wireless equipment shall use the channel and band assigned by the organizers. Personal hotspots and unauthorized transmitters are not permitted.

1.3.2 Teams may use a tether if it is managed by a designated tether handler and it never supports the rover's weight or aids traction.

1.3.3 The rover must be able to join the organizers' network for the ping task (the Software & Autonomous Rulebook). The network details are issued at the technical briefing.

1.3.4 In Autonomous Mode, the rover may transmit telemetry and video to the operator station, but no control command may flow from the operator to the rover.

1.4 Inspection

1.4.1 Every rover passes technical inspection before its first run. The inspection covers size, configuration (no arm or gripper), E-stop, power safety and communications.

1.4.2 Teams shall declare their **class** (Section 4) and any autonomous capabilities at inspection.

1.4.3 A rover that is modified after inspection must be re-inspected before running again.

1.4.4 Inspection also checks the mechanical and electrical design requirements of Sections 2 and 3. A rover that breaks a **mandatory** requirement is not permitted to run until it is corrected.

2. MECHANICAL DESIGN REQUIREMENTS

Mechanical design is documented and checked. The items below earn **Build Quality marks** in the General Rulebook (Build Quality Score).

2.1 CAD Design

2.1.1 The rover **shall be designed in CAD software** (for example Fusion 360, SolidWorks, FreeCAD, Onshape or Inventor) before it is built. Hand sketches alone are not accepted from Milestone M1 onward.

2.1.2 The CAD model shows the rover in its starting configuration inside the $300 \times 300 \times 300$ mm gauge, the flipper range of motion (Class B and C), the battery bay, and the position of the E-stop.

2.1.3 All design files shall be kept in the team's GitHub repository: the native CAD files, exported STEP or STL files, and a PDF or image of the full assembly.

2.1.4 The built rover must match the CAD model. Changes made during the build shall be reflected in the repository before the M4 freeze.

2.2 Fabrication and Assembly

2.2.1 **3D printing is preferred** for the chassis and brackets. Tough materials such as PETG, ABS or nylon are recommended; PLA should be used only for non-structural parts.

2.2.2 Other fabrication methods (laser-cut acrylic, aluminium, plywood and similar) are allowed, provided the CAD model exists and matches the rover.

2.2.3 Structural joints shall use bolts with lock nuts or threaded inserts. Hot glue and tape are not structural fasteners.

2.2.4 The battery bay and the E-stop must be rigidly mounted. Nothing on the rover may be loose, dangling or able to detach during a run.

2.2.5 Sharp edges and pinch points shall be rounded or guarded.

2.2.6 Teams shall be able to show their printable files (STL) at inspection on request.

3. ELECTRICAL REQUIREMENTS

All electronics must be built to a clean, safe and documented standard. Items marked **mandatory** are checked at technical inspection; the same items earn Build Quality marks at the M4 Build Review.

Topic	Required (mandatory)	Not permitted
Circuit construction	Soldered veroboard (stripboard or perfboard) or a PCB.	Breadboards of any kind; twisted or taped joints.
Main power connectors	XT connectors (XT30, XT60, XT90) or T (Deans) connectors.	Bare wires, alligator clips, tape or twisted joints on power lines.
Wire colours	Red for every positive power line, black for ground (GND).	Using red or black for signal wires.
Circuit design	A schematic made in circuit software, or a neat pen-and-paper diagram, with every pin labelled.	A circuit that is undocumented or differs from its schematic.

Table 2. Electrical requirements at a glance.

3.1 Circuit Construction

3.1.1 All circuits shall be built on **soldered veroboard** (stripboard or perfboard) or on a custom PCB. Joints must be clean, shiny and free of cold solder.

3.1.2 **Breadboards are not permitted** anywhere on the rover, including for temporary or test circuits during a run. A rover that carries a breadboard is not permitted to run.

3.1.3 Plug-in controller and sensor modules may sit in *soldered* female headers so they can be replaced. Every other connection shall be soldered or made with a proper crimped connector.

3.1.4 Wires shall be insulated (heat shrink or equivalent), tied down with cable ties or clips, and given strain relief at every connector. No exposed conductors.

3.1.5 Wire gauge shall be suitable for the current it carries, and motor wiring shall be kept apart from signal wiring where possible.

3.2 Power Connections and Wire Colours

3.2.1 Every power connection between the battery, the E-stop, the power distribution and the motor drivers shall use an **XT connector or T connector**. Bare wire, alligator clips and twisted or taped joints are not permitted on power lines.

3.2.2 Battery connectors must be polarised so that they cannot be plugged in reversed.

3.2.3 Wires shall follow the colour code in Table 3. Power and ground must be easy to tell apart at a glance.

Wire colour	Use
Red	Every positive power line: battery, 12 V, 5 V, 3.3 V. Label the voltage at each junction.
Black	Ground (GND) and battery negative.
Any other colour	Signal wires (for example yellow, green, blue, white). Never red or black.

Table 3. Wire colour code.

3.3 Power Source and Protection

3.3.1 All power is carried on board. The nominal battery voltage shall not exceed 24 V.

3.3.2 Batteries must be fused and secured against vibration and impact. LiPo and Li-ion packs must be carried and charged in fire-resistant bags.

3.3.3 Damaged, swollen or unlabelled batteries are not permitted in the pit or in the arena.

3.4 Circuit Design and Documentation

3.4.1 Every circuit shall be designed before it is built, either in **circuit-simulation or schematic software** (for example Proteus, KiCad, EasyEDA, Tinkercad Circuits, Fritzing or LTspice) or as a neat **pen-and-paper circuit diagram**.

3.4.2 The schematic shows the controller, drivers, sensors, motors, battery, fuse and E-stop, with **every pin mentioned**: controller pin numbers, driver and sensor pins, connector pins and the voltage of each power rail.

3.4.3 The schematic is submitted to the GitHub repository as a file (software export, PDF) or a clear photo of the hand-drawn diagram, together with the pin map.

3.4.4 The built circuit must match the schematic. If the circuit changes, the schematic must be updated before the M4 freeze.

4. ROBOT CLASSES

4.1 Classes and Class Multipliers

Rovers compete in one of three classes depending on their mobility hardware. The class multiplier is applied to *every* task a rover scores and rewards the added mechanical complexity of flippers.

Class	Mobility configuration	Class multiplier
Class A	Four-wheel drive (4WD); no flippers.	×1
Class B	Drive base with two flippers (front or rear pair).	×2
Class C	Drive base with four flippers.	×4

Table 4. Rover classes and class multipliers.

4.1.1 Flippers must be powered, controllable articulated limbs. Fixed, passive or spring-loaded extensions do not count as flippers.

4.1.2 For Class B and Class C a flipper must be able to rotate at least $120°$ and shall be used by the rover during the run, otherwise the Head Referee may reclassify the rover.

4.1.3 A team enters exactly one rover class per championship. Entering a rover in a lower class than it qualifies for is allowed but does not provide any extra bonus.

4.1.4 Classes compete in a single common ranking; the class multiplier is what balances the field.

5. SAFETY

5.1 Safety Requirements

5.1.1 Safety overrides all other rules. Any referee or official may stop a run at any time by calling “Stop”; the operator must immediately press the E-stop.

5.1.2 All team members in the arena and pit area must wear closed shoes. Eye protection is required when working with batteries, cutting tools or soldering.

5.1.3 Rovers must be transported with the E-stop pressed or power disconnected.

5.1.4 Only the technical team may carry out repairs in designated pit areas. Charging of batteries is only allowed in the charging area under supervision.

5.1.5 Sharp edges, exposed wiring and unprotected high-torque parts must be covered or guarded.

5.1.6 Spectators must stay behind the arena barriers.